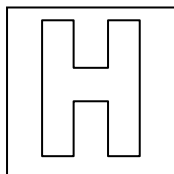


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PIONEER JUNIOR COLLEGE

2013 JC2 Preliminary Examination
HIGHER 1

CHEMISTRY
Paper 1

8872/01

26 September 2013

Additional Material: Data Booklet
 Multiple Choice Answer Sheet

50 minutes

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

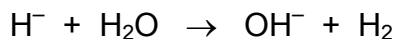
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 Group I and II ionic hydrides react with water:



In an experiment, 1 g samples of each of the following four ionic hydrides are treated with an excess of water.

Which sample produces the greatest volume of hydrogen gas?

- | | |
|-------------------------|-------------------------|
| A MgH_2 | B CaH_2 |
| C LiH | D NaH |

- 2 A 25 cm^3 sample of a 0.1 mol dm^{-3} ASO_4 solution requires 10 cm^3 of 0.1 mol dm^{-3} of acidified potassium manganate(VII) solution for complete reaction.

What is the final oxidation state of element **A**?

- | | |
|-------------|-------------|
| A +3 | B +4 |
| C +5 | D +6 |

- 3 Which electronic configuration represents an atom of an element, **E**, that forms a simple ion, E^{2-} ?

- A** $1s^2 2s^2 2p^6 3s^2 3p^2$
B $1s^2 2s^2 2p^6 3s^2 3p^4$
C $1s^2 2s^2 2p^6 3s^2 3p^6$
D $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

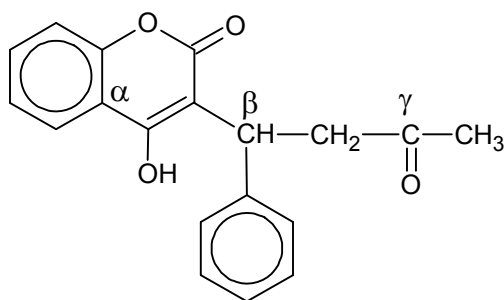
- 4 Which of the following sets consists of three compounds with different types of structure?

- A** P_4O_{10} , Na_2O , SiO_2
B AlCl_3 , Cu_2O , SiCl_4
C AlF_3 , MgCl_2 , BN
D BeO , SO_2 , BH_3

- 5 Which statement on the chemical properties of the oxides of the third period of the Periodic Table is true?
- A P_4O_{10} reacts with water to give OH^- .
- B Al_2O_3 is soluble in both $NaOH$ and HCl .
- C SiO_2 forms a solution of pH 6 when dissolved in water at room temperature.
- D Na_2O and MgO can be mixed in water to give an approximately neutral solution.
- 6 An element **J** does not react with cold water. However **J** reacts vigorously with oxygen to give an oxide which reacts readily with water forming an acidic solution.

What is element **J**?

- A magnesium
- B aluminium
- C silicon
- D phosphorus
- 7 *Warfarin* is used as a rat poison.

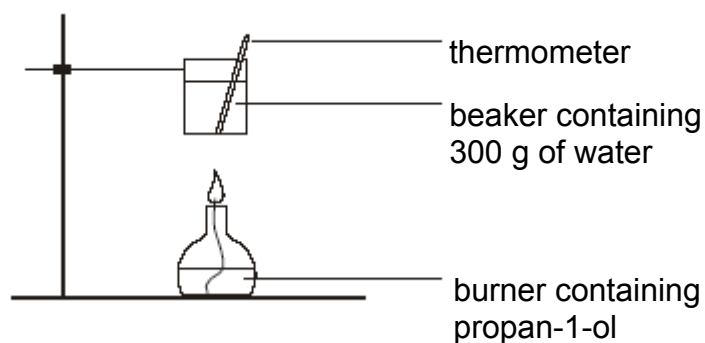


Warfarin

Which of the following gives the hybridisation of the carbon atoms at positions α , β and γ respectively?

	α	β	γ
A	sp^2	sp	sp
B	sp^2	sp^3	sp^2
C	sp^3	sp^2	sp
D	sp^3	sp^3	sp^3

- 8 A student used the apparatus below to heat a beaker containing 300 g of water.



The following data were recorded:

Mass of propan-1-ol burnt	= m g
Change in temperature of water	= ΔT °C

Given that:

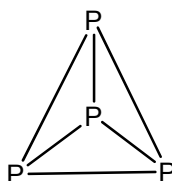
Relative molecular mass of propan-1-ol	= 60.0
Enthalpy change of combustion of propan-1-ol	= -2021 kJ mol ⁻¹
Specific heat capacity of water	= c kJ kg ⁻¹ K ⁻¹

What is the efficiency of this heating process?

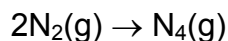
- A $\frac{m \times 2021 \times 1000}{300 \times c \times \Delta T \times 60.0} \times 100\%$
- B $\frac{300 \times c \times \Delta T \times 60.0}{m \times 2021} \times 100\%$
- C $\frac{m \times c \times \Delta T \times 60.0}{300 \times 2021 \times 1000} \times 100\%$
- D $\frac{300 \times c \times \Delta T \times 60.0}{m \times 2021 \times 1000} \times 100\%$

- 9 Use of the *Data Booklet* is relevant to this question.

Phosphorus, P₄, has the following molecular structure:



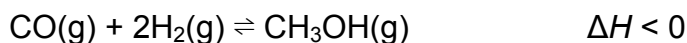
Assume that nitrogen were to form a similar molecule, N₄.



By considering the bonds broken and the bonds formed, what would be the value of ΔH for the above reaction?

- A 1028 B 1328 C 1954 D 2628

- 10 The following equilibrium exists in a system containing carbon monoxide and hydrogen gases.



Which of the following conditions would result in an increase in amount of methanol produced?

- A adding a catalyst
- B heating the system
- C adding of carbon monoxide
- D lowering the pressure of the system

- 11 Given that



it follows that, for the reaction,



What would be the value of X?

- | | |
|-------------------|--------------------------|
| A $\frac{1}{4.8}$ | B 2.4 |
| C $\frac{1}{2.4}$ | D $\frac{1}{\sqrt{4.8}}$ |

- 12 In which reaction is ammonia acting as an acid?

- A $2\text{NH}_3\text{(l)} + 2\text{Na(s)} \rightarrow 2\text{NaNH}_2\text{(s)} + \text{H}_2\text{(g)}$
- B $2\text{NH}_3\text{(g)} + 3\text{CuO(s)} \rightarrow 3\text{Cu(s)} + \text{N}_2\text{(g)} + 3\text{H}_2\text{O(g)}$
- C $\text{NH}_3\text{(g)} + \text{HCl(g)} \rightarrow \text{NH}_4\text{Cl(s)}$
- D $\text{NH}_3\text{(g)} + \text{H}_2\text{O(l)} \rightleftharpoons \text{NH}_4^+\text{(aq)} + \text{OH}^-\text{(aq)}$

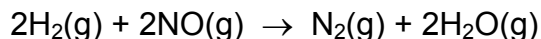
- 13 The rate equation for the reaction $\text{W} + 2\text{X} + \text{Y} \rightarrow \text{Z}$ is given below:

$$\text{Rate} = k[\text{X}]^2[\text{Y}]$$

If the concentration of **W** is tripled, the concentration of **X** is halved and the concentration of **Y** is doubled, by what factor is the rate of reaction being changed?

- | | | | |
|-------|-------|-------|-------|
| A 0.5 | B 1.0 | C 2.0 | D 4.0 |
|-------|-------|-------|-------|

- 14 Hydrogen and nitrogen monoxide can react to form nitrogen and steam.



The rate of reaction is first order with respect to hydrogen and second order with respect to nitrogen monoxide.

0.100 mol dm⁻³ of H₂ and 5.00 mol dm⁻³ of NO were put into a sealed flask of fixed volume. It was found that there was 0.0250 mol dm⁻³ of H₂ left 40 minutes later.

Which of the following statement is correct?

- A An excess concentration of 5.00 mol dm⁻³ NO(g) is used so that the rate of reaction will remain constant as reaction proceeds.
 - B There was 0.0250 mol dm⁻³ of N₂(g) formed after 20 minutes.
 - C There was 0.0125 mol dm⁻³ of H₂(g) left after 80 minutes.
 - D The unit of the rate constant is s⁻¹.
- 15 Alkanes undergo substitution with Cl₂ in the presence of ultraviolet light.

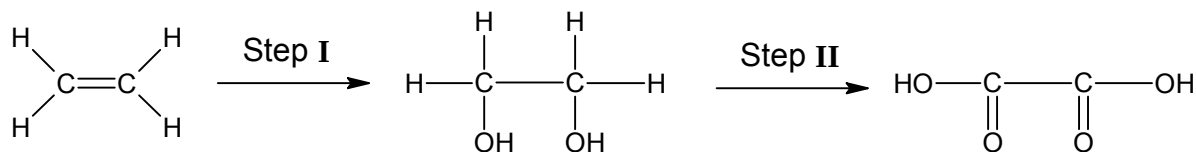
Which of the following could have been a possible starting material for the synthesis of 2-chloro-2-methylpentane?

- A CH₃CH₂CH₂CH₂CH₃
 - B CH₃CH(CH₃)CH₂CH₂CH₃
 - C CH₃CH₂CH(CH₃)CH₂CH₃
 - D CH₃CH₂CH₂CH₂CH₂CH₃
- 16 Bromine in an inert solvent is added separately to hexane, hexene, benzene and methylbenzene in the presence of sunlight.

Which of the following will **not** give the same observation as the others?

- A hexane
- B hexene
- C benzene
- D methylbenzene

17 Ethanedioic acid can be produced by the following steps:



Which reagents are used for steps I and II?

- | | Step I | Step II |
|----------|-------------------------------|---|
| A | cold alkaline KMnO_4 | acidified $\text{K}_2\text{Cr}_2\text{O}_7$, heat under reflux |
| B | cold alkaline KMnO_4 | acidified KMnO_4 , distil |
| C | hot alkaline KMnO_4 | acidified $\text{K}_2\text{Cr}_2\text{O}_7$, distil |
| D | hot alkaline KMnO_4 | acidified KMnO_4 , heat under reflux |
- 18 How many different isomeric alkenes, including geometric isomers, could be produced when $\text{C}_6\text{H}_5\text{CH}_2\text{CHBrCH}_3$ reacts with hot ethanolic KOH ?
- A** 3 **B** 4 **C** 5 **D** 6
- 19 Aqueous silver nitrate was added at the same time to separate solutions of chloroethane and iodoethane. The first signs of a reaction were in the sample containing iodoethane.

Why was the reaction with iodoethane noticed first?

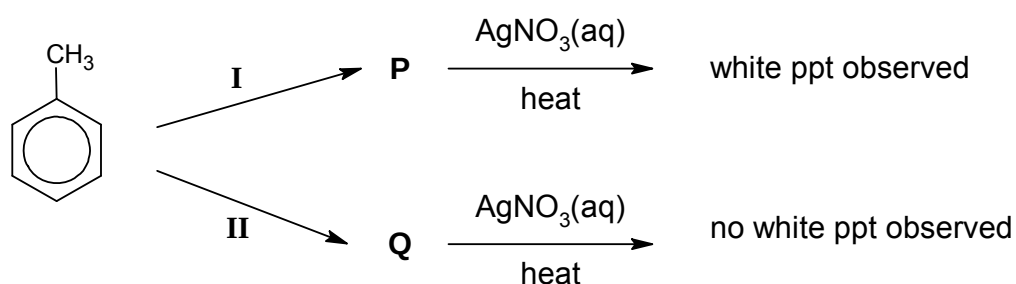
- A** The chloroethane also reacted with aqueous silver nitrate but gave a soluble product.
- B** The chloroethane reacted more slowly because the carbon-chlorine bond is less polar than the carbon-iodine bond.
- C** The iodoethane reacted more quickly because the carbon-iodine bond is weaker than the carbon-chlorine bond.
- D** The chloroethane reacted more slowly because the carbon-chlorine bond is longer than the carbon-iodine bond.

- 20 In 1911, Weizmann carried out a historically important conversion, changing starch into a mixture of butan-1-ol and propanone.

Which of the following reagents will **not** be able to distinguish the products of this conversion?

- A acidified manganate(VII) ions
- B 2,4-dinitrophenylhydrazine
- C alkaline aqueous iodine
- D Fehling's solution

- 21 Methylbenzene can be used as an additive in unleaded petrol. It can be reacted with chlorine in two ways, depending on the conditions of the reaction, as shown below.



Which of the following set of information is correct?

	I	P	II	Q
A	Cl_2 , uv		Cl_2 , Anhydrous AlCl_3	
B	Cl_2 , uv		Cl_2 , Anhydrous AlCl_3	
C	Cl_2 , Anhydrous AlCl_3		Cl_2 , uv	
D	Cl_2 , Anhydrous AlCl_3		Cl_2 , uv	

22 This question is about an organic compound **X**.

- When **X** was reacted with phosphorous pentachloride, fumes of hydrogen chloride were evolved.
- When **X** was warmed with acidified aqueous potassium dichromate(VI), the solution turned green.

Which of the following is a possible identity for **X**?

- | | |
|---|--|
| A $\text{CH}_3\text{CH}_2\text{CHO}$ | B $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ |
| C $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ | D CH_3COCH_3 |

23 Which sequence shows the correct order of increasing value of K_a ?

- A** $\text{CH}_3\text{CO}_2\text{H}$, $\text{C}/\text{CH}_2\text{CH}_2\text{OH}$, $\text{CH}_3\text{CH}_2\text{OH}$
B $\text{C}/\text{CH}_2\text{CO}_2\text{H}$, $\text{BrCH}_2\text{CO}_2\text{H}$, $\text{ICH}_2\text{CO}_2\text{H}$
C $\text{CH}_3\text{CHC}/\text{CO}_2\text{H}$, $\text{C}/\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$, $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
D $\text{Cl}_2\text{CHCH}_2\text{CO}_2\text{H}$, $\text{C}/\text{CH}_2\text{CHC}/\text{CO}_2\text{H}$, $\text{CH}_3\text{CCl}_2\text{CO}_2\text{H}$

24 Ester **Y**, which is used in perfumes, has the molecular formula $\text{C}_6\text{H}_{12}\text{O}_2$. It is found that one of the products of acid hydrolysis of **Y** gives a yellow precipitate with hot aqueous alkaline iodine.

What is the structural formula of **Y**?

- | | |
|--|---|
| A $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$ | B $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}_3$ |
| C $(\text{CH}_3)_2\text{CHCO}_2\text{CH}_2\text{CH}_3$ | D $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CO}_2\text{CH}_3$ |

25 A compound **W** with molecular formula $\text{C}_5\text{H}_8\text{O}_3$ has the following properties.

- It gives an orange precipitate with 2,4-dinitrophenylhydrazine
- It gives a yellow precipitate when warmed with alkaline aqueous iodine
- It does not react with Fehling's solution
- It reacts with Na_2CO_3 to liberate a gas which gives a white precipitate with limewater.

Which of the following is a possible identity for **W**?

- A** $\text{CHOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
B $\text{CHOCH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
C $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CO}_2\text{H}$
D $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CO}_2\text{H}$

Section B

For the next five questions, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The response **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct.	1 and 2 only are correct.	2 and 3 only are correct.	1 only is correct.

No other combination of statements is used as a correct response.

26 The table below shows the lattice energies of two compounds.

Compound	Lattice energy / kJ mol ⁻¹
NaF	-915
MgO	-3933

Which of the following statements help to explain the difference between these two values?

- 1 In each of these compounds, the ions are isoelectronic.
- 2 The electrostatic force of attraction between doubly charged ions is about four times that between singly charged ions.
- 3 The interionic distance in NaF is larger than that in MgO.

27 The reaction shown is reversible.



Which of the following will decrease when pressure is increased while temperature is kept constant?

- 1 The proportion of NO₂(g) present at equilibrium.
- 2 The value of the equilibrium constant K_c .
- 3 The activation energies of both forward and reverse reactions.

The response **A** to **D** should be selected on the basis of

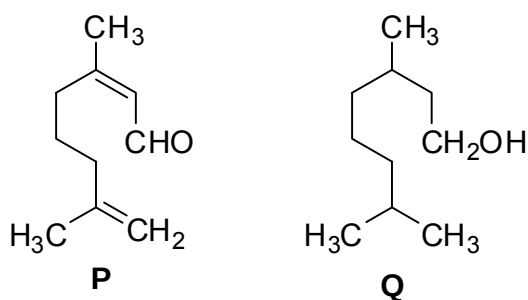
A	B	C	D
1, 2 and 3 are correct.	1 and 2 only are correct.	2 and 3 only are correct.	1 only is correct.

No other combination of statements is used as a correct response.

28 Which set of solutions of equal volume, when mixed, will **not** give an acidic buffer?

- 1** 0.10 mol dm⁻³ CH₃CO₂H and 0.10 mol dm⁻³ NaOH
- 2** 0.10 mol dm⁻³ CH₃CO₂H and 0.05 mol dm⁻³ Ca(OH)₂
- 3** 0.05 mol dm⁻³ HCl and 0.05 mol dm⁻³ CH₃CO₂Na

29 Compounds **P** and **Q** are important flavours in citrus fruits. These compounds are commonly used in the food and perfume industries.



Which statements concerning **P** and **Q** are correct?

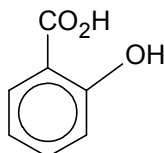
- 1** **Q** can be distinguished from **P** by using sodium metal.
- 2** **P** can be converted to **Q** using lithium aluminium hydride in dry ether.
- 3** Reaction of **P** with hot acidified KMnO₄ gives two organic products.

The response **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct.	1 and 2 only are correct.	2 and 3 only are correct.	1 only is correct.

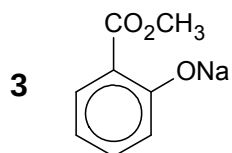
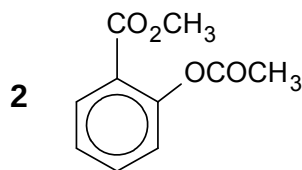
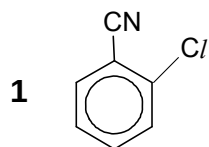
No other combination of statements is used as a correct response.

30 The diagram below shows the structure of salicyclic acid.



salicyclic acid

Which compounds give salicyclic acid on acid hydrolysis?



Answers

1	C	11	D	21	B
2	B	12	A	22	C
3	B	13	A	23	D
4	A	14	B	24	C
5	B	15	B	25	D
6	D	16	C	26	C
7	B	17	A	27	D
8	D	18	A	28	A
9	A	19	C	29	D
10	C	20	D	30	C

7 A

7 B

8 C

8 D